

ROS 2 Minimal C++ Package - Build and Common Fixes

This note shows the smallest possible C++ package in ROS 2, how to build it, and how to fix the most common problems. Everything is copy-paste friendly and uses only ASCII text.

Workspace Setup

What and why: A ROS 2 workspace is a folder where packages live and build together using colcon. Creating ~/ros2_ws/src sets a clean place to put your new package so that the build tools can find it.

```
# create workspace and src folder
mkdir -p ~/ros2_ws/src && cd ~/ros2_ws
# clone or create your package in src/
```

Build and Source

What and why: 'colcon build' compiles the workspace and installs artifacts into install/. Sourcing the setup files updates your environment so that ROS 2 can discover packages and executables.

```
cd ~/ros2_ws
colcon build --symlink-install
source /opt/ros/$ROS_DISTRO/setup.bash
source install/setup.bash
# optional (append once)
# echo "source ~/ros2_ws/install/setup.bash" >> ~/.bashrc
```

Minimal C++ Package

What and why: package.xml declares metadata and dependencies so the build system knows what to pull in. CMakeLists.txt describes how to build your node and where to install the executable so 'ros2 run' can find it.

```
# package.xml
<package format="3">
  <name>your_pkg</name>
  <version>0.0.1</version>
  <description>Example</description>
  <maintainer email="you@example.com">You</maintainer>
  <license>Apache-2.0</license>
  <buildtool_depend>ament_cmake</buildtool_depend>
  <depend>rclcpp</depend>
</package>
# CMakeLists.txt
cmake_minimum_required(VERSION 3.8)
project(your_pkg)

find_package(ament_cmake REQUIRED)
find_package(rclcpp REQUIRED)

add_executable(talker src/talker.cpp)
target_compile_features(talker PUBLIC cxx_std_17)
ament_target_dependencies(talker rclcpp)

install(TARGETS talker
  RUNTIME DESTINATION lib/${PROJECT_NAME})

ament_package()
```

Build and Run

What and why: Limiting the build to your_pkg keeps iteration fast. After sourcing, 'ros2 run your_pkg talker' executes the compiled binary that was installed under install/your_pkg/lib/your_pkg/.

```
cd ~/ros2_ws
colcon build --symlink-install --packages-select your_pkg
source install/setup.bash
ros2 run your_pkg talker
```

Common Fixes

What and why: Builds can break when headers or targets move, or when environment overlays are stale. These steps reset the workspace, verify what was installed, and validate that the environment is sourced correctly.

```
# Clean stale build artifacts
cd ~/ros2_ws
rm -rf build/ install/ log/
colcon build --symlink-install
source install/setup.bash

# Check installed targets
ls -l install/YOUR_PKG/lib/YOUR_PKG/
ros2 pkg executables YOUR_PKG

# Ensure names match correctly
# - Package name matches <name> in package.xml
# - Executable name matches target name in CMakeLists.txt

# Overlay sanity shell
env -i bash --noprofile --norc -c '
  source /opt/ros/$ROS_DISTRO/setup.bash;
  source ~/ros2_ws/install/setup.bash;
  ros2 pkg list | wc -l'
```